Data Mining Psychological Diagnoses to Identify Vulnerable Populations in Chihuahua, Mexico

Project Checkpoint

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ABSTRACT

Mental health vulnerability in Chihuahua, Mexico, the state with the highest suicide rate in the country, remains poorly understood despite its high ranking on well-being indices. This project mines 18,220 psychological consultation records from the Chihuahua city government to identify demographic and diagnostic patterns among help-seekers. After cleaning and standardizing 30+ inconsistent diagnosis labels, 20+ occupations, and education categories, the analysis combined population comparison (age, sex, education), Cramér's V correlation analysis, ordinary least squares regression of monthly consultations on temperature, and k-prototypes clustering of mixed-type demographic data with UMAP projection. Key findings: (1) the sample is 60.2% female and strongly bimodal in age, with peaks at 15–19 and 45–54; (2) help-seekers are dramatically over-represented in ages 10–19 relative to the Chihuahua population, most extremely for males; (3) temperature shows no significant association with monthly consultation volume, and (5) six interpretable demographic clusters emerge, dominated by adolescents with anxiety/ADHD and middle-aged adults with grief or partner conflict. These patterns suggest that help-seeking is concentrated in a much younger and more female population than the state average, pointing to both unmet need among adults and the potential value of school-based mental health interventions.

1 INTRODUCTION

Chihuahua city is the capital of the homonym state, located in northern Mexico and bordering the US. Although it is one of the top states in the Better Life Index, it is also the one with the highest suicide rate. The rate in the country is 6.4 per 100,000 persons, while in Chihuahua it more than doubles to 13.7 per 100,000. This signals a hidden vulnerability, where although the general population appears healthy on aggregate indicators, specific subgroups may be disproportionately vulnerable. This project analyzes 18,220 psychological diagnoses of Chihuahua's citizens to identify demographic and diagnostic patterns among help-seekers. Contribution: this work provides a demographic comparison of help-seekers against the Chihuahua population and a mixed-type clustering of consultations that yields interpretable vulnerable subgroups. Together these results inform where mental health resources should be prioritized in a state where aggregate prosperity masks hidden distress.



Figure 1: Mexico, Chihuahua highlighted. Map source: Wikimedia Commons

2 RELATED WORK

There have been few studies about mental health in Chihuahua, specifically for suicide. One study by Oropeza-Acevedo in 2025 looked at patients admitted at a Chihuahua hospital for self-harm injuries. The investigation analyzed trends such as sobriety, physical illnesses and psychiatric disorders. The last of these should be differentiated from psychological diagnosis, as while disorders describe symptoms or behaviors, on the record of psychological diagnosis a source of origin for the problem is also usually included. Therefore, we can use this previous information to compare whether the same kind of groups suffering from suicidal ideation are the ones seeking professional psychology attention, and where differences lie.

The second study by Fernández-López et al. measures the seasonality of suicide, finding that each 1°C rise in temperature was associated with a 2.8% increase in the suicide rate from 2008 to 2018. Since our chosen database includes month of consultation we could also determine if there is an increase in

patients seeking help along these lines, by attaching historical temperature data to our chosen database.

Our results extend Fernández-López et al. in two ways. First, we do not find a statistically significant association between monthly temperature and consultation volume ($p = 0.56$), suggesting that the seasonal signal documented for completed suicides does not directly translate into help-seeking behavior. Second, the demographic concentration of help-seekers (adolescents and middle-aged adults) differs from the demographic profile of suicide attempters documented by Oropeza-Acevedo.

3 PROPOSED WORK

The objective of this project is to mine psychological consultations and diagnoses of Chihuahua's citizens to yield patterns. Some of the results we found are vulnerable population clusters that can be targeted as a priority for the allotment of resources.

3.1 Dataset

The dataset consists of 18,220 psychological consultation records obtained from the Chihuahua city government under the freedom of information act. Each record contains patient age, sex, education level, occupation, marital status, neighborhood, consultation month and year, and assigned psychological diagnosis.

3.2 Preprocessing

All categorical fields were recoded from Spanish to English. Education was mapped to five ordered levels (none, primary, secondary, high school, bachelor); small categories (kindergarten, post-doc) were merged into adjacent levels to avoid unstable estimates. Occupation was consolidated from 20+ raw labels into 11 semantic categories (student, employed, self-employed, professional, healthcare, service, retired, unemployed, homemaker, and a residual category). Diagnosis was standardized from 30+ inconsistent Spanish strings (with accents and typos) into 28 canonical English labels. Duplicate entries and records with missing demographic fields were retained in the master table but excluded from models requiring complete cases.

3.3 Tools

Analysis was conducted in R (Positron IDE) using tidyverse for wrangling, plotly for visualization, clustMixType for mixed-type clustering, uwot for UMAP projection, and cluster for Gower distance computation. Reference population data was obtained from Data México's population pyramid for Chihuahua.

3.4 Analytical Techniques

- **Population comparison.** Sample age and sex distribution was binned to match the 18 population age ranges and compared via a population pyramid. Education distribution was compared against census percentages using a chi-square goodness-of-fit test.
- **Correlation analysis.** Cramér's V was used for pairwise association between categorical variables.
- **Seasonality.** Monthly consultation counts were regressed on historical monthly average temperatures from Metar-Taf.com using ordinary least squares.
- Clustering. k-prototypes was chosen over k-means because the feature space mixes numeric (age) and categorical (gender, education, occupation, marital status) variables. Gower distance and UMAP were used for 2D projection. Silhouette scores guided k selection.

4 EVALUATION

4.1 Metrics

Statistical significance is assessed at $\alpha = 0.05$. Because analyses are exploratory, pairwise post-hoc comparisons use Bonferroni adjustment. For the education comparison, we report both χ^2 and Cramér's V, since the very large sample size makes statistical significance trivially easy to obtain and effect size more informative. Clustering is validated with silhouette scores across k 3 to 10, and cluster structure is inspected with a UMAP projection on Gower distances. Reproducibility is ensured by a fixed random seed (42) throughout.

4.2 Experimental setup

Population comparisons use the Data México pyramid for Chihuahua and published census education proportions. The seasonal model uses 18 monthly observations (Jan 2025 to Jun 2026) matched to historical mean temperatures from Metar-Taf.com.

5 RESULTS

5.1 Demographic Profile

The sample is 60.2% female (n = 10,963) and 38.9% male (n = 7,094), with 163 unspecified (Figure 2). The age distribution is clearly bimodal (Figure 3), with a dominant peak at 15–19 and a secondary bulge at 45–54, separated by a dip among working-age adults (25–40). Education is dominated by secondary (n = 6,239) and high school (n = 4,652), with only 44 unspecified cases (Figure 4).

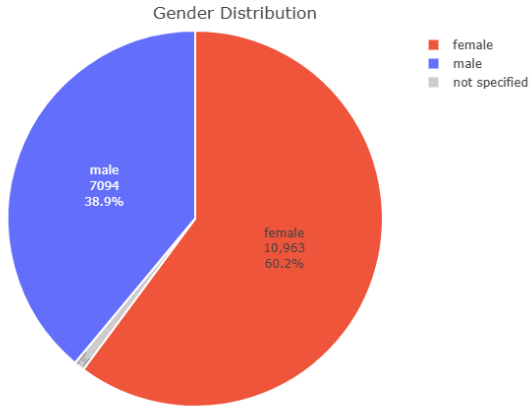


Figure 2: Gender Distribution.

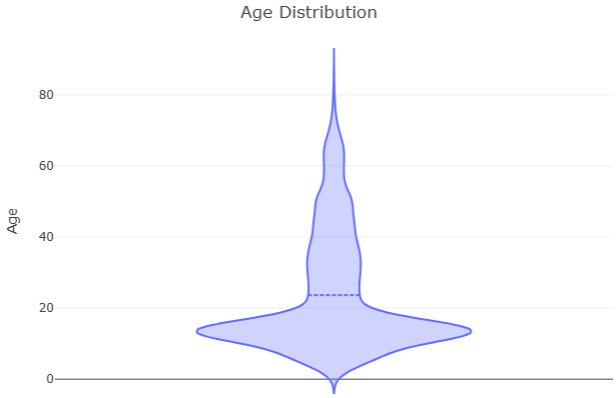


Figure 3: Age Distribution.

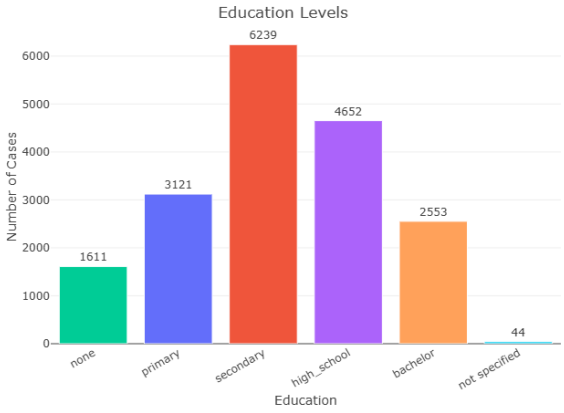
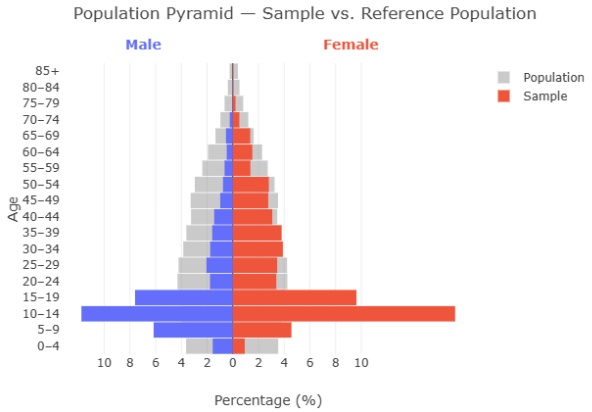


Figure 4: Educational Level Distribution.

5.2 Comparison with the Reference Population

The population pyramid (Figure 5) reveals substantial over-representation of help-seekers in the 10 to 19 age range for both sexes, most dramatically for males aged 10 to 19. Adults aged 25 to 49 are under-represented. The female sample peaks at 10



to 19; the male sample peaks even more sharply at 10 to 14. These patterns indicate that help-seeking is concentrated at younger ages than the population at large.

Figure 5: Sample vs Reference Population

5.3 Correlation Structure

The Cramér's V matrix (Figure 6) shows only modest associations between categorical variables. The strongest is occupation against marital status (V = 0.28), followed by occupation against education (V = 0.27) and diagnosis against education (V = 0.26). Gender is essentially independent of all other variables (max V = 0.22 with diagnosis). This supports treating these features as largely orthogonal inputs to clustering.

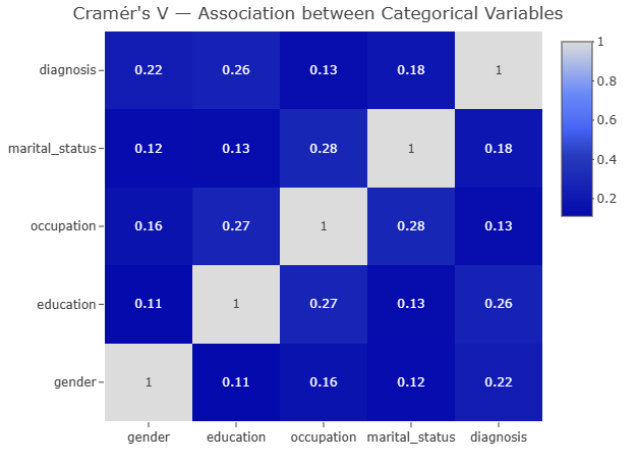


Figure 6: Correlation Matrix.

5.4 Seasonality and Temperature

Monthly consultation counts regressed on average monthly temperature yield a non-significant slope ($\beta = -13.4$, SE = 22.6, $p = 0.56$). Temperature does not appear to predict help-seeking volume in this sample, in contrast with published seasonality findings for completed suicides.

5.5 Clustering

k-prototypes with k = 6 was fitted on age, gender, education, occupation, and marital status. Silhouette scores peaked at this k (0.37). The UMAP projection (Figure 7) shows the six clusters with modest but visible separation. Cluster profiles reveal distinct demographic profiles:

1. Middle-aged people struggling with grief: Mean age 52, mixed gender, dominant diagnosis grief.
2. Kids struggling with ADHD: mean age 9, mixed gender, dominant diagnosis adhd.
3. Male highschoolers struggling with Anxiety: mean age 17, majority male, dominant diagnosis generalized anxiety.
4. Secondary students struggling with Depression: mean age 13, dominant education secondary, dominant diagnosis depression.
5. Female highschoolers struggling with Depression: mean age 17, majority female, dominant diagnosis depression.
6. Adults struggling with Partner Conflict: Mean age 37, mixed gender, dominant diagnosis conflictual relationship with spouse or partner.

These six profiles suggest that help-seeking divides primarily along two axes: life stage (child, adolescent, adult) and primary presenting concern (anxiety, depression, ADHD, grief, relationship conflict).



Figure 7: UMAP cluster representation.

6 TIMELINE AND DISCUSSION

5.1 Timeline

Stage	Status	Notes
Project proposal	Done	
Data mining	Done	18,220 records from Chihuahua city open data portal.
Data preprocessing	Done	30+ diagnosis labels, 20+ occupations, and education recoded; merging small education categories.
Data exploration	Done	Descriptive statistics, violin plot, pie chart, population pyramid.
Correlation & hypothesis testing	Done	Cramér's V matrix; chi-square education comparison.
Clustering	Done	k-prototypes (k = 6) + UMAP projection.
Visualizations	Done	All figures finalized.
Final report and Presentation	Done	Project completed October 2nd.

5.2 Changes from the proposal

Three substantive methodology changes were made:

- **k-means to k-prototypes.** The feature space is predominantly categorical; Euclidean distance on one-hot encodings would be dominated by high-cardinality fields (e.g., occupation, neighborhood). k-prototypes handles mixed types natively via a weighted combination of Gower-style distance and Euclidean distance.
- **PCA to UMAP on Gower distances.** PCA assumes continuous features and produces a misleading projection for mixed data. UMAP on Gower distances preserves the local structure that k-prototypes actually clusters on.
- **Seasonal decomposition to linear regression on monthly counts.** With only 18 monthly observations (Jan 2025 – Jun 2026), time-series decomposition is statistically underpowered. OLS on temperature provides a clean, interpretable test of the seasonality hypothesis within the data's limits.

5.3 Challenges encountered

- **Inconsistent diagnosis labels.** The raw diagnosis field contained 30+ Spanish strings with accent, capitalization, and spacing variations. A systematic recoding pass and validation against the original strings was required.

- **Rare education categories.** kindergarten and post-doc had very small counts and were merged into none and bachelor, respectively, to avoid unstable estimates in the chi-square and clustering.
- **Age binning.** The reference population uses 18 age ranges; matching required careful binning of the continuous age field and handling of boundary values.
- **k-means failed.** Initial k-means attempts on one-hot encoded data produced poorly separated clusters, motivating the switch to k-prototypes.

5.4 Lessons learned

- **Data type drives method choice.** The initial k-means plan was reasonable in the abstract, but the data's predominantly categorical nature made k-prototypes the correct choice once the feature space was understood.
- **Effect size matters more than p-value at scale.** With n = 18,220, nearly any difference is statistically significant. Reporting Cramér's V alongside χ^2 prevents over-interpreting small effects.
- **Negative results are results.** The non-significant temperature effect is itself informative: it suggests the seasonality pattern documented for completed suicides does not extend to help-seeking behavior, which is a meaningful finding for public health planning.

CONCLUSION

This project mined 18,220 psychological consultation records from Chihuahua, Mexico, to identify demographic and diagnostic patterns among help-seekers. The dataset was cleaned, recoded, and analyzed through population comparison, correlation analysis, seasonal regression, and mixed-type clustering. Findings indicate that help-seekers in Chihuahua are disproportionately young (peak at 15–19), disproportionately female (60.2%), and that their education distribution differs significantly from the general population. Temperature shows no significant association with consultation volume in this sample. Six interpretable demographic clusters emerge, defined primarily by life stage (child, adolescent, adult) and presenting concern (anxiety, depression, ADHD, grief, partner conflict), identified via k-prototypes and projected with UMAP.

These findings suggest that, in a state where aggregate prosperity masks severe psychological distress, help-seeking is concentrated in a much younger and more female population than the state average. This points to both the value of school-based interventions and the possibility of unmet need among working-age adults.

Future work. There are multiple ways to strengthen this analysis. First, augment the data with emergency-room records, suicide-attempt registries, or school-based mental health reports, this would help capture populations that never reach the public psychology system and address the self-selection limitation directly. Second, scale the analysis geographically comparing Chihuahua to other low suicide rate Mexican states would help isolate whether the demographic patterns observed here are state-specific or general to northern Mexico.

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